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Intelligent Tuition Reimbursement Agent Proposed Generative AI Solution by ThinkMindLabs

Document Classification: **CONFIDENTIAL — CLIENT USE ONLY**

Prepared for: Human Resource, Montefiore Health System

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Document Version: v1.0 | April 2026

Programs Covered:

- Regular Tuition Reimbursement — All Employees (incl. NYSNA)
- CME Reimbursement — PAs and Non-Union Physical Therapists
- MMC Scholarship — Children of Montefiore Employees
- Dependent Tuition — Children of Physicians, Scientists & Executives

Note. Assumption of policy, amounts, values and remunerations are assumptions used as placeholders to create example user journeys.

1. Executive Summary

Executive Summary

Montefiore Health System is one of New York's premier academic health systems, serving the Bronx, Westchester, and Hudson Valley communities with a workforce spanning nurses, physicians, scientists, executives, and administrative staff. Today, Montefiore manages four distinct tuition and scholarship benefit programs covering multiple cases per year. Despite the strategic importance of these benefits—both as workforce retention tools and as commitments under collectiveagreements—the current process is almost entirely manual. *Source: (open world internet)*

Based on our understanding and discovery of the current position at hand, most submissions require multiple manual touchpoints due to missing or incorrect documentation. Processing relies on email chains, paper forms, ServiceNow comments, and HR staff manually verifying eligibility, validating transcripts, chasing signatures, and re-keying approved amounts into Workday payroll. There is no audit trail, no real-time visibility for employees, and no automated enforcement of critical policy rules—including the NYSNA 45-day reimbursement deadline and the service agreement requirement for amounts over

This document presents ThinkMindLabs' proposed Generative Agentic AI solution: the Montefiore Tuition Reimbursement Agent (MTRA). The MTRA is an ensemble of eight purpose-built AI agents that work collaboratively to automate the end-to-end lifecycle of every tuition benefit program—from initial eligibility verification through document ingestion, approval orchestration, payroll integration, compliance monitoring, and audit logging.

The solution is designed to meet the stringent compliance requirements of U.S. healthcare software, including HIPAA, SOC 2 Type II, NIST 800-53, and applicable New York State data protection regulations. It is deployable as SaaS, PaaS, or On-Premise, and integrates natively with Montefiore's existing ServiceNow environment, Workday HRIS and Payroll, and Microsoft Teams—requiring no disruption to the systems your workforce already uses.

What This Solution Delivers

- End-to-end automation of all four tuition programs with AI agents handling intake, validation, routing, and payroll posting
- Elimination of the 50%+ re-work rate through proactive missing-document detection and real-time employee guidance
- NYSNA 45-day and service agreement compliance enforced automatically—no HR manual tracking required

- Real-time status visibility for employees, managers, and HR through a unified portal or native Workday / MS Teams experience
- Immutable audit log and compliance dashboard ready for internal, union, and regulatory review at any time
- AI governance architecture aligned with Montefiore's IT Security requirements and U.S. healthcare compliance standards
- Extensible agent framework designed to accommodate new programs, policy changes, and expanded automation over time

2. Solution Overview

Solution Overview

2.1 The Montefiore Tuition Reimbursement Agent (MTRA)

The MTRA is a generative agentic AI system built on a multi-agent orchestration framework. Unlike conventional automation (which follows rigid, pre-scripted decision trees), the MTRA uses large language model (LLM) reasoning combined with domain-specific rules engines to handle the full range of structured and unstructured decisions that characterize healthcare HR workflows: reading and interpreting transcripts, understanding service agreement terms, recognizing policy exceptions, and composing contextual communications to employees in plain language.

Each agent in the ensemble is a self-contained AI module with a defined purpose, data inputs, decision authority, and escalation threshold. Agents collaborate through an Orchestration Layer that manages task sequencing, parallel execution, inter-agent data sharing, and exception routing. No single agent makes a consequential decision in isolation—every output is logged, explainable, and auditable.

2.2 Design Principles

Human-in-the-loop by design	AI agents handle all repetitive, rule-based, and data-intensive tasks. Human judgment is preserved for policy exceptions, final scholarship awards, and high-value approvals. All AI auto-decisions can be reviewed and overridden by HR.
Explainability first	Every agent decision generates a human-readable explanation stored in the case record. HR staff see not just what the agent decided, but why—enabling confident review and confident override when appropriate.
Policy-as-code	All tuition program rules (NYSNA 45-day window, $3K service agreement threshold, 18-credit annual cap, IRS §127 tax treatment) are encoded as version-controlled policy objects—not hard-coded logic. Policy changes are made by HR administrators without developer involvement.
Zero-trust data architecture	No employee PII or benefit data leaves Montefiore's defined data boundary without explicit encryption and access control. Agents operate on tokenized

	identifiers where possible, accessing source systems via secured API integration.
Compliance-native	HIPAA Privacy and Security Rules, SOC 2 Type II, NIST 800-53, and New York SHIELD Act requirements are embedded in the architecture—not bolted on after the fact. Every design decision is evaluated against these standards.
Extensible agent registry	The agent framework is built to accommodate new programs (e.g., loan forgiveness tracking, certification reimbursements) and new automation capabilities without architectural rework. Montefiore owns the roadmap.

2.3 Programs and Policy Rules Encoded in the System

Based on the discovery session conducted on 29 April 2026, the following programs and their specific policy constraints are encoded into the MTRA agent ruleset:

Program	Eligible Population	Annual Cap	Key Rules	Taxability
Regular Tuition + NYSNA	All employees incl. NYSNA members	$3,000 / 18 credits p.a.	45-day NYSNA reimbursement window; service agreement if >$3K	IRS §127 up to $5,250 non-taxable
CME Reimbursement	PAs and Non-Union Physical Therapists	Per department budget	PTs are unionized—union-specific rule review required; AMA PRA Category validation	Program-specific; varies
MMC Scholarship	Children of Montefiore employees (need-based)	$500 per award cycle	Financial need criteria; employee tenure ≥3 years; committee award decision	Generally non-taxable scholarship
Dependent Tuition (Docs/Scientists/Execs)	Children of attending physicians, executives, principal scientists	$4,000–$6,000 p.a.	Accredited institution verification (IPEDS); dependent relationship	Taxable above IRS §127 exclusion; W-2 impact

			verification; CHRO approval required	

Confirmed Rigid Policy Constraints — Encoded as Non-Negotiable Rules

- NYSNA 45-day reimbursement deadline: enforced by automated SLA tracking; HR escalated if at risk
- Service Agreement mandatory for all reimbursements exceeding $3,000: digital generation, e-signature routing, and lifecycle monitoring fully automated
- 18 credits per academic year cap: enforced at intake; cumulative YTD credit tracking maintained per employee
- Credit limit (18 credits) identified by client as the single highest source of confusion and exceptions today—AI agent provides proactive balance alerts

3. Users & Process Owner Mapping

Users & Process Owner Mapping

A clear understanding of who uses the system, who owns the current process, and where the greatest pain sits is foundational to designing an AI agent that delivers real value. This section maps every user group, their current responsibilities, their pain points, and the AI agent or capability that directly addresses their experience. This mapping informed the design of every agent in the MTRA.

3.1 Two Distinct Experience Layers

The MTRA serves two fundamentally different user experiences. Conflating them produces a system that serves neither well. This distinction drives every UX and agent design decision.

Frontend Layer — Employee Self-Service	Backend Layer — Operations & Control
Who: All Montefiore employees across all four programs: non-union staff, NYSNA nurses, PAs, PTs, physicians, executives, scientists, and employees applying on behalf of dependent children. **What they need:** Simplicity. Guided step-by-step application with plain-language instructions. Real-time feedback on what is missing. Status visibility throughout. Policy rules enforced invisibly by the system. **Design principle:** Zero HR contact required for a clean, complete submission. The Submission Assistant Agent guides the employee from program selection to final document upload without any HR intervention.	**Who:** HR / Tuition Operations Team, Department Approvers (Managers, Dept Admins, Dept Heads), Payroll Team, and Compliance / Audit Stakeholders. **What they need:** Automation and control. Maximum reduction of manual work. AI-generated recommendations with full override capability. Exception management. Real-time compliance visibility. No manual data re-entry between systems. **Design principle:** AI agents handle 80–85% of case processing end-to-end. HR touches only genuine exceptions and high-value approvals. Every AI action is explainable and overridable by an authorized human.

3.2 Persona 1: Employees (Primary Applicants — Frontend Users)

This group encompasses all Montefiore staff who submit tuition or scholarship benefit applications. They are the primary end users of the self-service portal and the primary beneficiaries of the automated experience.

Sub-Segment	Current Responsibilities	Programs Accessed
Non-union employees **RNs / NYSNA union members** **Physician Assistants (PAs)** **Non-union Physical Therapists** **Physicians, Executives, Scientists**	Submit tuition or scholarship applications Upload supporting documents (receipts, transcripts, acceptance letters, aid statements, service agreements) Track application status after submission Respond to missing information requests from HR Sign service agreements (currently paper-based and manual)	Own tuition reimbursement (all employees + NYSNA) — $3,000 / 18 credits CME reimbursement (PAs and non-union PTs) MMC Scholarship for dependent children (all employees, need-based, $500 cap) Dependent tuition for children of Physicians, Executives, Scientists ($4,000–$6,000)

Current Pain Points	What MTRA Changes for This Persona
Confusing and inconsistent requirements across the four programs No single submission channel—email, paper forms, and ServiceNow used inconsistently Multiple re-submissions required due to missing documents (affects >50% of all cases today) No real-time visibility into application status after submission 18-credit annual cap confusing—employees frequently unaware of remaining balance before submitting Service agreement process is paper-based, causing delays and document loss risk	One unified portal for all four programs with guided, step-by-step intake and plain-language instructions Real-time eligibility check and credit/dollar balance displayed before document upload begins Instant gap report after upload—specific missing items listed with exact requirements, not generic errors Real-time status tracker visible at all times; automated status updates at each workflow stage Digital service agreement generated and sent for e-signature automatically—no paper process Payment confirmed in next paycheck cycle after all documents received and approved

3.3 Persona 2: HR / Tuition Operations Team (Core Process Owners — Backend)

This team is the primary internal operator and carries the highest manual workload today. They are the principal target for AI-driven automation within the MTRA. Every current manual step they perform has a corresponding MTRA agent that replaces or augments it.

Key Insight: This Group Is the Central Target for AI Automation

- The HR Tuition Team currently performs tasks that are ideally suited to AI: reading documents, checking grades, verifying credits, following up on missing files, preparing payroll spreadsheets, and maintaining paper records.
- Under the MTRA, HR Specialists review AI-prepared case summaries and make decisions only on genuine exceptions. They no longer read documents, check policy tables, or chase employees for documents.
- A clean, complete submission requires zero HR intervention from intake through payroll posting. HR touches only the estimated 5–10% of cases that are genuine exceptions or policy edge-cases.
- ~1,000 cases per year at 4–6 hours each = an estimated 10,000–15,000 HR hours per year currently consumed by tuition administration. MTRA target: < 2,000 hours per year (exception handling only).

Current Manual Responsibilities	Current Pain Points
Review every application and all supporting documents manually	Fully manual document reading—no OCR, no extraction tools in current system
Validate grades (C or higher), eligible expenses, and policy compliance (credit caps, dollar limits)	>50% of all cases require multiple manual follow-up cycles
Follow up with employees on missing or incorrect documents (highest-volume step)	Payroll spreadsheet preparation is error-prone and creates additional processing delays
Store documents and maintain employee records across paper and digital systems	No formal audit trail exists beyond email threads and paper files
Manually prepare reimbursement data in payroll spreadsheets for Workday handoff	Inconsistent processing across programs and campuses due to union status differences
Track NYSNA 45-day compliance and service agreement status via manual spreadsheet	No compliance dashboard—audit readiness requires 40–60 hours of manual preparation

3.4 Persona 3: Approvers — Managers, Dept Admins & Dept Heads

Approvers are not daily system users, but their timely action is critical to case progression. Today they represent a significant bottleneck: approval chains are non-standard, there are no SLAs, and the entire process relies on paper signatures or ad-hoc email exchanges.

Who They Are	Department Managers (direct line managers—but not always the approver per client confirmation) | Department Administrators (in some departments this role approves, not the manager) | Department Heads (senior approver for higher-value cases) | CHRO / HR Director (mandatory final approver for Program 4) | Finance VP (for reimbursements exceeding $5,000)
Current Challenges	Approval chains are non-standard and vary by department. No defined SLAs—cases sit unreviewed with no escalation mechanism. Paper-based signature process creates a physical bottleneck. Approvers receive no structured case summary. No visibility into whether their approval is time-sensitive (e.g., NYSNA 45-day window at risk).
MTRA Experience	Receives an AI-prepared structured case summary via DocuSign email link or MS Teams Adaptive Card. Summary displays: employee name, program, amount, AI eligibility result, document validation status, and policy compliance. One-click Approve or Decline. Decline requires a comment field. Automatic 24-hour escalation if no action taken. NYSNA urgency flag shown on time-sensitive cases. CHRO delegation enforced automatically if unavailable.

3.5 Persona 4: Payroll Team

The Payroll Team is the final step before payment reaches the employee. Today they receive a manually prepared spreadsheet from HR and re-key data into Workday payroll. The MTRA eliminates this handoff entirely through direct API integration.

Current Responsibility	Current Pain	MTRA Future State	Residual Role Under MTRA
Receive reimbursement spreadsheet from HR Tuition Team. Re-key amounts and GL codes into Workday. Verify tax treatment manually. Process payment in next available payroll cycle.	Manual data re-entry is error-prone. HR spreadsheet preparation adds 5–10 day delay. Tax treatment for Program 4 dependent tuition calculated manually and inconsistently. No automated GL coding.	Payroll Integration Agent posts approved amounts directly to Workday via API. No spreadsheet. Tax treatment pre-calculated by Eligibility Agent. GL code from department configuration. Repayment deductions also automated on SA breach.	Payroll Team receives automated posting notification with case reference, amount, and GL code for reconciliation. Exception handling and W-2 year-end review remain human. Payroll processing volume does not change—method changes.

3.6 Persona 5: Compliance & Audit Stakeholders

Compliance and audit stakeholders are not daily system users, but their requirements are critical architectural constraints. Montefiore confirmed no formal audit process is currently in place—but the organization must be fully audit-ready. The MTRA makes audit readiness a continuous, zero-effort state rather than a periodic labor-intensive exercise.

Stakeholder	Current State	MTRA Provides
HR Director / CHROInternal AuditLegal & Compliance	No compliance dashboard. Audit data compiled manually from emails, paper files, and ServiceNow comments. No anomaly detection or exception flagging.	Real-time compliance dashboard showing: $ spent by program, role, employee type, school, degree, and dependent cases. Immutable audit log. Anomaly and exception alerts. On-demand report generation in under 1 minute. 10-year retention enforced automatically.
NYSNA Union RepresentativesLabor Relations	NYSNA compliance tracked manually via spreadsheet. 45-day window adherence at risk. No automated enforcement or union-specific reporting.	NYSNA-specific compliance report generated automatically: 45-day window adherence rate, service agreement status per employee, reimbursement amounts by CBA category. Available on-demand at any time for Labor Relations review.
Finance / CFO OfficeTax & Payroll Compliance	IRS §127 taxable/non-taxable split calculated manually. W-2 implications tracked via HR spreadsheet. No annual reconciliation report generated.	Tax Compliance Agent calculates IRS §127 split at point of intake for every applicable case. Annual reconciliation report generated automatically for W-2 preparation and tax filing. Finance receives real-time spend dashboard.

3.7 External & Indirect Participants

These parties are not system users but influence the document inputs and workflow. The MTRA accounts for their role without requiring them to access the system directly.

Educational Institutions	Provide official transcripts and tuition receipts that employees upload to the portal. The Document Intake Agent processes these documents regardless of formatting variation across institutions. For Program 4, institutional accreditation is verified automatically via IPEDS/DOE API at submission.
Notary (Service Agreements)	Currently required for paper service agreement notarization in some cases. Under the MTRA, service agreements are executed via DocuSign e-signature—legally equivalent under the U.S. ESIGN Act and New York Electronic Signatures and Records Act (ESRA). Standard service agreement notarization requirement is eliminated. Montefiore Legal confirms where notarization remains legally required for specific agreement types.
College Board / Financial Aid Offices	Provide financial aid statements and FAFSA-related documentation for the MMC Scholarship program ($500 need-based). Employees upload these as part of the digital application. Document Agent classifies and validates financial need documentation format automatically.

3.8 Complete Process Owner Mapping

The following table maps every step of the current tuition reimbursement process to its current owner, the associated pain point, and the MTRA agent that takes ownership in the future state.

Process Step	Current Owner	Current Pain Point	MTRA Agent	Future Mode
Application intake & program selection	**Employee**	No guided experience; confusing requirements; multiple submission channels (email, paper, ServiceNow)	**Submission Assistant (Document Intake Agent)**	**Fully Automated**
Eligibility verification (tenure, FTE, union)	**HR Tuition Team**	Manual HRIS lookup; 30–60 min per case; inconsistently applied across HR staff	**Eligibility & Policy Compliance Agent**	**Fully Automated**
18-credit cap & dollar limit check	**HR Tuition Team**	Manual spreadsheet calculation; 18-credit limit is the #1 source of employee confusion and exceptions	**Eligibility & Policy Compliance Agent**	**Fully Automated**

Document collection from employees	**Employee + HR Tuition Team**	>50% of cases have missing documents; no structured checklist provided upfront to employees	**Missing Document Detection Agent**	**Fully Automated**
Document reading & field extraction	**HR Tuition Team**	Fully manual; 4–6 hours per case; the single highest HR workload step in the process	**Document Intake & Extraction Agent**	**Fully Automated**
Grade validation (C+ general; B+ NYSNA grad)	**HR Tuition Team**	Manual transcript reading; grade threshold inconsistently applied across HR staff	**Document Intake & Extraction Agent**	**Fully Automated**
CME certificate classification	**Dept Admin / HR**	Decentralized across departments; AMA PRA Category 1 verification inconsistent	**Document Intake Agent (CME mode)**	**Fully Automated**
Service agreement generation & routing	**HR Tuition Team**	Paper form; frequently lost; no lifecycle tracking; no expiration monitoring whatsoever	**Service Agreement Lifecycle Agent**	**Fully Automated**
Approval routing (manager/dept head)	**HR Tuition Team**	Sequential and blocking; paper or ad-hoc email; no SLAs; no escalation; variable chains by dept	**Approval Routing & Signature Agent**	**Auto routing; Human decision**
Follow-up on missing documents	**HR Tuition Team**	Manual emails; most time-consuming step; >50% of cases require multiple follow-up cycles	**Notification & Communication Agent**	**Fully Automated**
NYSNA 45-day compliance monitoring	**HR Tuition Team**	Manual spreadsheet tracking; deadline frequently at risk with no early-warning system	**SA Lifecycle Agent + Notification Agent**	**Fully Automated**

Service agreement expiration tracking	**HR Tuition Team**	Paper files; no automated alerts; breaches discovered only after the fact	**Service Agreement Lifecycle Agent**	**Fully Automated**
Repayment calculation on early departure	**HR Tuition Team**	Manual proration calculation; inconsistently applied; frequently disputed by departing employees	**Service Agreement Lifecycle Agent**	**Fully Automated**
Payroll spreadsheet preparation	**HR Tuition Team + Payroll Team**	Manual data re-entry across two teams; error-prone; adds 5–10 business days to payment timeline	**Payroll Integration Agent**	**Fully Automated**
Payroll posting to Workday	**Payroll Team**	Re-keying approved amounts; manual GL coding; tax treatment calculated by hand	**Payroll Integration Agent**	**Fully Automated**
IRS §127 tax split calculation	**HR / Payroll / Finance**	Manual; inconsistently applied; no automated W-2 impact tracking for Program 4	**Eligibility Agent — Tax Compliance function**	**Fully Automated**
Scholarship committee scoring	**HR + Committee Members**	No structured rubric in any system; paper/email scoring; no conflict-of-interest detection	**Scholarship Scoring Module (Document Agent)**	**AI-Assisted; Human Decision**
Audit log maintenance	**None (no formal process today)**	No formal audit trail beyond email threads; significant manual prep required for any review	**Audit, Compliance & Reporting Agent**	**Fully Automated**
Compliance & union reporting	**HR Tuition Team (manual)**	40–60 hours of preparation required per audit; no real-time dashboard available	**Audit, Compliance & Reporting Agent**	**Fully Automated**

10-year data retention enforcement	**HR / IT (ad-hoc)**	No automated retention policy; manual archiving inconsistent across programs and locations	**Audit, Compliance & Reporting Agent**	**Fully Automated**

3.9 Agent-to-Persona Mapping

Each MTRA agent is purpose-mapped to the persona it primarily serves, ensuring design, output format, and escalation behavior are calibrated to the specific needs of that user group.

MTRA Agent	Primary Persona Served	How It Serves That Persona
Submission Assistant(Document Intake Agent — Employee Mode)	**Employees (all programs)**	Guides employee through structured intake; pre-populates known fields from HRIS; validates documents instantly with AI; shows actionable gap report in plain language; eliminates confusion about what each program requires
Eligibility & Policy Compliance Agent	**Employees + HR Tuition Team**	Displays eligibility status and remaining credit/dollar balance to employees before upload. Removes HR eligibility verification entirely for standard cases. Enforces 18-credit cap, dollar limits, NYSNA rules, and IRS §127 tax treatment automatically.
Missing Document Detection Agent	**Employees + HR Tuition Team**	Employees receive specific, actionable gap reports instead of generic HR emails. HR no longer reviews incomplete submissions—agent handles the entire re-submission detection and notification cycle automatically.
Notification & Communication Agent	**All Personas**	Employees: real-time status updates and proactive deadline reminders. Managers: structured approval requests with auto-escalation. HR: case action alerts and NYSNA deadline warnings. Payroll: posting confirmations. Compliance: exception and anomaly alerts.
Approval Routing & Signature Agent	**Approvers: Managers, Dept Admins, Dept Heads, CHRO**	Delivers AI-prepared case summary via DocuSign email or MS Teams Adaptive Card. One-click Approve / Decline. Automatic 24-hour escalation

		if no action. Non-standard approval chains configured per department by HR Admin without developer support.
Service Agreement Lifecycle Agent	**HR Tuition Team + NYSNA / Labor Relations + Payroll**	Generates service agreement digitally and routes for e-signature. Monitors commitment period. Alerts HR before expiration. Calculates prorated repayment automatically on departure event. Eliminates all manual SA tracking by the HR Tuition Team.
Payroll Integration Agent	**Payroll Team + HR Tuition Team**	Eliminates the payroll spreadsheet entirely. Posts directly to Workday via API. Applies pre-calculated tax treatment. Codes GL accounts from department configuration. Processes repayment deductions on SA breach. Payroll Team role shifts to reconciliation only.
Audit, Compliance & Reporting Agent	**Compliance Stakeholders + HR Director + NYSNA Representatives + Finance**	Maintains continuous immutable audit log. Provides real-time compliance dashboard. Generates NYSNA-specific reports on demand. Produces IRS §127 annual reconciliation report. Sends exception alerts. Audit readiness is a permanent zero-effort state—not a periodic preparation exercise.

4. The AI Agent Ecosystem

The AI Agent Ecosystem

The MTRA comprises eight specialized AI agents, each with clearly defined responsibilities, data authority, and escalation behavior. Together they form a closed-loop automation system that requires human intervention only for policy exceptions, high-value approvals, and final award decisions where human judgment is both appropriate and required.

📄 Agent 1: Document Intake & Extraction Agent	
Purpose	Receives, classifies, and extracts structured data from all submitted documents across all four programs. Replaces 100% of manual HR document reading for standard submissions.
Key Inputs	Uploaded files (PDF, JPG, PNG, TIFF): transcripts, tuition receipts, acceptance letters, tuition aid statements, service agreements, CME certificates, dependent verification documents
Outputs	Structured JSON record with all extracted fields (institution, course, grade, credit hours, amount, dates, accreditor, degree level); document classification; validation result (PASS / FAIL / REVIEW); confidence score (0–100); field-level confidence breakdown
Automation Level	Fully Automated. OCR engine (Tesseract / AWS Textract) extracts raw text; LLM (GPT-4 class) interprets and structures fields; domain-specific validation rules applied per program type. Human review triggered only when confidence < 85% or critical field missing.

✅ Agent 2: Eligibility & Policy Compliance Agent	

Purpose	Real-time eligibility determination at the moment of application submission. Enforces all program-specific rules including credit caps, dollar limits, tenure requirements, NYSNA membership, and IRS §127 tax treatment for Program 4.
Key Inputs	Employee record from Workday HRIS (tenure, FTE status, job classification, union code, active employment); program selection; application data; YTD claims history
Outputs	Eligibility determination (ELIGIBLE / INELIGIBLE / NEEDS_REVIEW); specific reason codes; applied limit and remaining balance; NYSNA membership confirmation; IRS §127 taxable/non-taxable split calculation for dependent tuition program; policy flags
Automation Level	Fully Automated via real-time HRIS API query. Determination returned in under 30 seconds at point of intake. Configurable rule engine—HR Administrators can update policy parameters without developer support.

🔍 **Agent 3: Missing Document Detection Agent**	
Purpose	Proactively identifies gaps in submitted documentation before and after submission. Eliminates the reactive, manual follow-up cycle that currently affects over 50% of all cases.
Key Inputs	Program-specific required document checklist; submitted document records from Document Intake Agent; validation results; application metadata
Outputs	Gap report listing each missing or insufficient document with specific description of what is required; personalized employee notification via ServiceNow and email; case status update; re-submission tracking
Automation Level	Fully Automated. Gap check runs within 30 minutes of submission. Employee receives actionable, plain-language notification within 1 hour. Agent monitors re-submissions and re-runs validation automatically—no HR involvement required for standard gaps.

📨 Agent 4: Notification & Communication Agent	
Purpose	Manages all proactive and reactive communications across the entire application lifecycle for all user types: employees, managers, HR specialists, Finance, and the NYSNA compliance team.
Key Inputs	Application status events; calendar triggers (45-day NYSNA deadline, 90-day completion deadline, academic year reset, service agreement expiration); employee profile and communication preferences; notification templates
Outputs	Personalized multi-channel notifications (ServiceNow, email; SMS capability available) at each lifecycle trigger. Communications reference the employee by name, cite the specific program, remaining balance, and deadline. All outbound communications logged to case record.
Automation Level	Fully Automated. Notification engine operates on event triggers from all other agents. Message personalization via LLM-generated plain-language content. All templates are rewritten as part of implementation (client confirmed templates to be rewritten). No HR manual drafting required.

✏️ Agent 5: Approval Routing & Signature Agent	
Purpose	Orchestrates parallel digital approval routing, signature collection (service agreements), and escalation logic—eliminating the sequential, blocking approval chains that create delays today.
Key Inputs	Approved case data; approval chain configuration (manager, Dept Admin, Dept Head—variable by department per client confirmation); DocuSign / Adobe Sign API; delegation matrix; dollar thresholds
Outputs	Parallel approval requests dispatched simultaneously (not sequentially); auto-escalation after configurable timeout (24 hrs manager, 48 hrs HR); CHRO escalation for high-value dependent tuition cases; service agreement generated, routed for e-signature, and filed

	to employee record on completion; all signature events logged with timestamp and signatory identity
Automation Level	Automated routing and escalation. Approval decisions remain human. The agent eliminates wait time from routing delays but does not replace human judgment at the decision point. Average signature collection target: 1–3 business days vs. current 8–15 days.

📋 Agent 6: Service Agreement Lifecycle Agent	
Purpose	Manages the full lifecycle of RN and employee service agreements—from generation through monitoring, expiration alerting, and automated repayment calculation on early departure. Specifically designed to enforce the $3,000 service agreement rule confirmed in discovery.
Key Inputs	Service agreement terms; employee departure events from Workday (termination trigger); agreement expiration dates; reimbursement amounts and disbursement dates
Outputs	Digital service agreement generated per case; expiration alerts dispatched at 90 / 60 / 30 days before end of commitment; on departure event: prorated repayment amount calculated automatically; repayment workflow initiated; HR and Finance notified with full calculation breakdown
Automation Level	Fully Automated monitoring and calculation. HR receives exception alert only on departure event requiring repayment action. Repayment formula: (Reimbursement Amount × Remaining Months ÷ Total Commitment Months). Repayment deduction posted to Workday payroll via Payroll Agent.

💳 Agent 7: Payroll Integration Agent	

Purpose	Automates the posting of all approved reimbursement amounts directly to Workday payroll, eliminating manual re-keying, accelerating payment to the next available payroll cycle, and calculating correct tax treatment.
Key Inputs	Approved case record with reimbursement amount, employee ID, program type, GL cost center; IRS §127 tax classification from Eligibility Agent; Workday Payroll API
Outputs	Payroll posting confirmation; GL entry with department cost center coding; tax withholding calculation applied (non-taxable portion excluded under IRS §127; taxable fringe benefit portion processed with correct withholding for Program 4); payment date communicated to employee; case closed in ServiceNow
Automation Level	Fully Automated. Payment posted within 4 business hours of final approval. Target: payment in next available payroll cycle (client confirmed goal: payment within next paycheck after all documents received). W-2 implications updated for Program 4 gross-up cases.

📊 **Agent 8: Audit, Compliance & Reporting Agent**	
Purpose	Maintains a continuous, immutable audit log of every action across all agents and human actors. Generates compliance dashboards and reports for HR leadership, Finance, Legal, Internal Audit, and NYSNA union representatives.
Key Inputs	Event feeds from all seven agents; human action logs (approvals, overrides, exceptions); case metadata; financial data; policy configuration history
Outputs	Real-time compliance dashboard ($ spent by program, role, employee type, school, degree level, dependent cases); immutable audit log with actor, timestamp, action, and decision rationale; NYSNA-specific compliance report (45-day window adherence, service agreement status); IRS §127 annual reconciliation report; exception and anomaly alerts; 10-year data retention (configurable per Montefiore policy)
Automation Level	Fully Automated logging and reporting. Dashboard updates in real-time. Monthly compliance reports generated automatically. Audit-ready at all times—no manual preparation required for HR or union review.

5. User Journeys

User Journeys

The following journeys illustrate how each user persona experiences the MTRA-powered system. Mode indicators reflect the nature of each step: Fully Automated (AI agents handle end-to-end), AI-Assisted (AI prepares the work; human decides), and Human Decision (judgment required).

4.1 Journey: Employee — Regular Tuition Reimbursement (Program 1)

Persona: Any Montefiore employee (non-union or NYSNA member) seeking reimbursement for an eligible course up to $3,000 / 18 credits per academic year.

#	Step	Agent / System Action	Mode	SLA
1	Employee logs into self-service portal (or Workday / Teams)	Eligibility Agent queries Workday HRIS in real-time: verifies active employment, tenure, FTE status, YTD credit balance, annual dollar balance. Result displayed instantly on screen.	**Fully Automated**	< 30 seconds
2	Employee completes structured intake form—one unified form per program	Form fields are pre-populated from HRIS where available (employee name, department, cost center). Validation runs inline as employee types. Incomplete fields flagged before submission.	**Fully Automated**	Employee-paced
3	Employee uploads supporting documents (transcript, receipt, acceptance letter)	Document Intake Agent runs OCR + LLM extraction. Extracts: institution, course name, grade, credit hours, tuition amount. Grade threshold (C or better) validated automatically. Receipt line items parsed.	**Fully Automated**	< 15 minutes
4	Gap Report displayed immediately after upload	Missing Document Agent compares submitted set against program checklist. Missing items listed in plain language with specific instructions. Employee	**Fully Automated**	< 1 hour

		prompted to re-upload before submission is marked complete.		
5	Application submitted; employee receives confirmation with case ID	Notification Agent sends confirmation via ServiceNow and email with case ID, expected next steps, and balance impact. Employee can track real-time status at any time.	**Fully Automated**	Immediate
6	Manager receives parallel approval request	Approval Routing Agent dispatches DocuSign request to manager (or Dept Admin / Dept Head per department configuration). Auto-escalation after 24 hours of no response.	**Fully Automated routing; Human decision**	1–3 business days
7	HR Benefits Specialist reviews AI-validated case summary	HR sees: Eligibility Agent result, Document Agent extracted fields and confidence scores, gap report, policy compliance check. One-click approve or flag for exception. No manual document reading required for clean cases.	**AI-Assisted**	< 1 business day
8	Payroll Integration Agent posts approved amount	Amount posted to Workday payroll for next cycle. Tax treatment applied. Payment confirmation sent to employee with expected payment date.	**Fully Automated**	< 4 hours post-approval
9	Audit Agent closes case and archives all records	Immutable case record created with full history: every agent action, every human decision, every document, every notification. Retained per Montefiore data retention policy (10 years).	**Fully Automated**	Real-time

4.2 Journey: RN / NYSNA Member — Service Agreement Tuition (Program 1, NYSNA rules)

Persona: Registered Nurse covered by the NYSNA Collective Bargaining Agreement. Amount requested is over $3,000, triggering mandatory service agreement. NYSNA 45-day reimbursement window applies.

#	Step	Agent / System Action	Mode	SLA

1	RN logs in; selects tuition program; NYSNA status auto-detected	Eligibility Agent confirms NYSNA membership, tenure, FTE, and remaining NYSNA-specific annual balance. NYSNA-specific form variant and policy rules activated automatically.	**Fully Automated**	< 30 seconds
2	RN completes intake form; amount entered triggers service agreement requirement	System detects amount > $3,000. Service Agreement Lifecycle Agent generates personalized service agreement document referencing employee name, amount, commitment period, and repayment formula. Agreement delivered via DocuSign.	**Fully Automated**	< 5 minutes
3	RN reviews and e-signs service agreement	DocuSign routing completed. Signed agreement automatically filed to employee record in ServiceNow and Workday. Case status updated. 45-day NYSNA clock begins.	**Human Decision (signature)**	1–2 days
4	RN uploads documents (transcript, tuition receipt, acceptance letter)	Document Intake Agent applies NYSNA-specific grade threshold (B or better for graduate-level). Amount validated against NYSNA limit. All extracted fields logged.	**Fully Automated**	< 15 minutes
5	NYSNA-specific gap check runs	Missing Document Agent verifies signed SA is on file, grade threshold is met, program is nursing-related (per CBA). Any gap triggers targeted notification to RN with specific instructions.	**Fully Automated**	< 1 hour
6	HR Business Partner reviews with NYSNA compliance summary	HR receives AI-generated NYSNA compliance summary: 45-day window status, SA confirmation, grade validation, dollar limit check. One-click approve. If exception, Labor Relations automatically notified.	**AI-Assisted**	< 1 business day
7	45-day NYSNA compliance monitoring active throughout	Notification Agent sends escalation alerts to HR at 30 days, 10 days, and 5 days before NYSNA deadline. If at risk, NYSNA Compliance Officer receives alert.	**Fully Automated**	Continuous

8	Payroll posts reimbursement within next paycheck	Payroll Agent posts approved amount. NYSNA compliance report records successful reimbursement within 45-day window.	**Fully Automated**	Next payroll cycle
9	Service agreement lifecycle monitoring begins	SA Lifecycle Agent monitors commitment period. Alerts at 90 / 60 / 30 days before end. On departure: prorated repayment automatically calculated and HR notified.	**Fully Automated (ongoing)**	Full commitment period

4.3 Journey: PA / Physical Therapist — CME Reimbursement (Program 2)

Persona: Physician Assistant or Non-Union Physical Therapist seeking reimbursement for AMA PRA Category 1 CME activity. Note: PTs have a unionized component—the system flags any PT-specific union rule review requirements.

#	Step	Agent / System Action	Mode	SLA
1	Clinician submits CME request via unified portal or Workday	Eligibility Agent verifies provider type (PA vs. PT), union status flag for PTs, department CME budget availability, individual annual limit.	**Fully Automated**	< 30 seconds
2	CME certificate and itemized conference receipts uploaded	Document Agent classifies certificate type (AMA PRA Category 1 vs. other), extracts credit hours, accreditor name, and conference dates. Receipt Agent parses line items and flags non-reimbursable expenses (entertainment, alcohol, personal items).	**Fully Automated**	< 15 minutes
3	Real-time CME budget check displayed	Eligibility Agent shows current YTD spend, department budget utilization, and remaining individual limit. Over-limit warning displayed before submission if applicable.	**Fully Automated**	< 30 seconds
4	Department Admin receives AI-prepared approval summary	One-click approval interface with AI-extracted CME details, budget	**AI-Assisted**	< 1 business day

		impact, and certificate validation result. Replaces email-based admin review.		
5	Payment routed via correct channel	Payroll Agent determines correct payment method by provider type (payroll vs. Accounts Payable). Posts via appropriate channel. GL coded to department CME cost center.	**Fully Automated**	Next payroll or AP cycle
6	PT-specific union flag reviewed if applicable	For PT submissions, Compliance Agent flags for HR/Labor Relations review to confirm applicable union CME rules before approval. Ensures no CBA violations.	**AI-Assisted (PT only)**	Per HR review schedule

4.4 Journey: Employee Parent — MMC Scholarship Application (Program 3)

Persona: Montefiore employee with ≥3 years tenure applying for the need-based $500 scholarship for a dependent child.

#	Step	Agent / System Action	Mode	SLA
1	Scholarship portal opens 90 days before annual window; employee receives personalized invitation	Notification Agent queries HRIS for eligible employees (tenure ≥3 years with registered dependents). Sends personalized invitation 90 / 60 / 30 / 7 days before deadline via ServiceNow and email.	**Fully Automated**	90 days pre-window
2	Employee submits digital scholarship application with dependent details	Eligibility Agent auto-verifies employee tenure and dependent relationship via Workday dependent records. Pre-populates known fields. Flags incomplete dependent records for manual verification.	**Fully Automated**	< 30 seconds
3	Financial need documentation submitted	Document Agent processes financial need evidence. Missing Document Agent checks completeness and notifies employee of any gaps.	**Fully Automated**	< 15 minutes

4	Scholarship committee receives structured scoring portal	Each committee member accesses a digital scoring rubric (financial need, academic standing, other criteria). AI generates a standardized, anonymized summary of each application. Conflict-of-interest flag raised if committee member has a direct reporting relationship with the employee.	**AI-Assisted**	Per committee schedule
5	Committee scores submitted; AI aggregates and ranks	Scoring Agent aggregates all member scores with statistical normalization. Ranked shortlist presented to HR Director for final review. Individual member scores kept private until all members submit.	**AI-Assisted**	Automated aggregation
6	HR Director makes final award decisions	Final human decision on awards. Award letters auto-generated and sent via portal and email. Non-awardees receive a respectful, personalized notification.	**Human Decision**	Per committee schedule
7	Finance Bot initiates ACH scholarship disbursement	Payment of $500 per awarded scholarship disbursed directly to verified educational institution via ACH. Confirmation sent to employee and institution. GL coded and case closed.	**Fully Automated**	< 2 business days post-decision

4.5 Journey: Physician / Executive — Dependent Tuition Reimbursement (Program 4)

Persona: Attending Physician, C-suite Executive, or Principal Scientist submitting dependent tuition reimbursement ($4,000–$6,000). IRS §127 tax treatment applies. CHRO or HR Director approval required.

#	Step	Agent / System Action	Mode	SLA
1	Physician/Executive accesses high-touch portal; job classification auto-verified	Eligibility Agent: pulls job classification from Workday to confirm eligible category (attending physician, C-suite, principal scientist). Active employment confirmed. Dependent relationship checked against Workday dependent records.	**Fully Automated**	< 30 seconds

2	Institution tuition invoice uploaded	Document Agent extracts institution name, tuition amount, and academic year from invoice. Compliance Agent queries IPEDS/DOE database in real-time to verify institutional accreditation.	**Fully Automated**	< 5 minutes
3	IRS §127 Tax Calculation Panel displayed to employee	Tax Compliance function calculates: total requested, non-taxable portion (IRS §127 exclusion, currently $5,250/year), taxable fringe benefit above threshold, estimated withholding impact. Labeled as an AI-calculated estimate; employee prompted to consult personal tax advisor for binding advice.	**Fully Automated**	< 30 seconds
4	Dependent relationship documentation reviewed	For dependent verification, employee uploads required documentation (birth certificate, tax dependent record). AI validates identity document completeness. Inconsistencies flagged for HR review.	**AI-Assisted**	< 30 min AI; 1 day HR
5	CHRO / HR Director approval routing	Approval Routing Agent dispatches to CHRO via DocuSign. If no response in 48 hours, auto-escalates to designated Deputy CHRO. Delegation policy enforced and logged. Finance VP also notified for amounts above IRS exclusion threshold.	**Automated routing; Human decision**	1–2 business days
6	Payroll Agent posts with correct tax treatment	Non-taxable portion processed as excluded. Taxable fringe benefit portion processed with appropriate withholding. W-2 implications recorded. Tax summary sent to employee.	**Fully Automated**	< 4 hours post-approval
7	IRS §127 annual reconciliation logged	Audit Agent records all data for annual IRS §127 reconciliation report. Supports accurate W-2 preparation at year-end.	**Fully Automated**	Ongoing + annual

4.6 Journey: HR Benefits Specialist — Daily Case Management

Persona: HR Benefits Specialist managing the full case queue across all four programs. This is the primary operational view for the Montefiore HR Tuition Team.

#	Step	Agent / System Action	Mode	SLA
1	HR logs into HR Dashboard; case queue displays all active cases	Queue shows all cases filtered by program, status, priority, SLA risk, and AI confidence score. NYSNA 45-day deadline countdown visible per relevant case. One-click access to any case detail.	**Fully Automated (display)**	Real-time
2	HR opens a case; sees AI Agent Output Panel	Structured panel shows: Eligibility Agent result, Document Intake Agent extracted fields + confidence scores, Missing Document Agent gap report, Service Agreement status, NYSNA compliance status. HR sees the full AI reasoning, not just a recommendation.	**AI-Assisted**	< 1 minute to review
3	HR approves clean case with one click	For cases where all agents return PASS and no flags are raised, HR reviews the AI summary and approves. No manual document reading. Payroll Agent triggered automatically.	**AI-Assisted; Human approval**	< 5 minutes per case
4	HR flags exception on complex case	For cases with REVIEW flags (low confidence, policy ambiguity, union exception), HR accesses original documents with agent annotations, overrides agent recommendation with documented reason, and routes to appropriate escalation.	**Human Decision**	Per complexity
5	HR manages NYSNA compliance dashboard	Real-time view of all NYSNA cases: 45-day window status, service agreement compliance, reimbursement amounts, outstanding signatures. Automated alerts already sent; HR has full visibility.	**AI-Assisted**	Continuous
6	HR generates union or leadership report on demand	Audit & Reporting Agent generates: NYSNA compliance summary, program spend by category, exception log, SLA adherence report. Available instantly—no manual data compilation.	**Fully Automated**	< 1 minute

6. Agent Integration Options

Agent Integration Options

The MTRA is designed for native integration with Montefiore's existing technology ecosystem. Rather than requiring a new standalone system that employees must learn and adopt separately, the agents can be surfaced through the tools Montefiore's workforce already uses. Three integration pathways are available and can be deployed simultaneously or in phased sequence.

5.1 Integration Option A: Dedicated Web Application

A purpose-built, responsive web application serves as the primary self-service portal for employees, HR staff, managers, and committee members. This is the richest integration option and serves as the foundation on which Workday and Teams integrations are layered.

Access method	Secure SSO login via Montefiore's identity provider (Azure AD / Okta). All roles access the same URL; views are role-differentiated at the application layer.
Employee experience	Program selection, AI-guided intake form, document upload with real-time AI validation status panel, gap report, application status tracker with timeline, notification center, benefit balance viewer.
HR experience	Case queue with filtering and sorting, AI Agent Output Panel per case, one-click approval, exception management, NYSNA compliance dashboard, service agreement tracker, audit log, on-demand reporting.
Manager experience	Email-link or SSO access to approval view. AI-prepared case summary. Approve / Decline with comment. Auto-escalation countdown visible.
Committee experience	Scholarship scoring portal. Anonymized applicant cards with AI summaries. Structured rubric. Private scoring. Aggregate ranked view after all scores submitted.
Mobile responsiveness	Full functionality on mobile browsers (375px viewport). Critical for nurses and clinicians who may submit on personal devices.

Technology stack	React 18 + TypeScript frontend. REST API backend (Java Spring Boot or Node.js). ServiceNow integration via REST API for case synchronization. FHIR-compatible data interfaces where applicable.
ServiceNow alignment	All cases created and updated in ServiceNow. The web application is a superior UX layer over ServiceNow—not a replacement. Internal goal of keeping all communications in ServiceNow is fully preserved.

5.2 Integration Option B: Workday Native Integration

For Montefiore staff who live in Workday for time management, payroll review, and HR self-service, the MTRA can be surfaced as a native Workday extension. This integration ensures the tuition benefit experience is within an environment employees already trust and use daily.

Integration approach	Workday Extend platform (Workday's native application development framework) or Workday Integration Cloud (WIC) for API-based connection. UI surfaces directly inside Workday HCM.
Employee experience	Tuition application accessible from Workday homepage or Benefits module. Pre-populated with Workday employee data. Document upload and status tracking within Workday interface.
HR / Payroll experience	Approved reimbursements posted directly to Workday Payroll via Workday REST API. Tax calculations written back to Workday for W-2 processing. No re-keying. HR can access case summaries within Workday.
HRIS data utilization	Eligibility Agent reads employee tenure, FTE status, job classification, union code, and dependent records directly from Workday. Dependent relationship verification uses Workday dependent data as primary source.
Service agreement data	Service agreements filed to Workday employee record as attachments. SA expiration dates written to Workday for lifecycle monitoring.
Limitations to note	Workday Extend has UI constraints relative to a custom web application. Full AI Agent Output Panel richness may require the web application as a companion for HR staff. Employee-facing flows are well-suited to Workday; HR operations flows are richer in the web application.

Recommended for	Organizations that want to consolidate employee self-service touchpoints within Workday. Ideal for the employee intake experience and payroll posting integration.

5.3 Integration Option C: Microsoft Teams Bot

Given Microsoft Teams' widespread adoption across healthcare organizations for internal communication, a Teams-native MTRA bot enables conversational interaction with the agent ecosystem without leaving the collaboration platform. This is particularly valuable for notification delivery, quick approvals, and status inquiries.

Integration approach	Microsoft Azure Bot Framework integrated with Teams via Microsoft Graph API and Adaptive Cards for rich, interactive message formats.
Employee experience	Employees can ask the MTRA bot natural language questions: "What is my tuition balance?", "What documents do I still need to submit?", "When will my reimbursement be paid?" Bot responds with real-time data from the agent ecosystem.
Manager approval	Managers receive Adaptive Card approval requests directly in Teams DM. Card shows AI case summary with Approve / Decline buttons and comment field. Response triggers case status update without leaving Teams.
HR notifications	HR Specialists receive Teams notifications for cases requiring attention: new cases assigned, NYSNA deadline alerts, exception flags. Deep-link to web application case detail included.
Conversational intake	Employees can initiate a tuition application through a guided Teams conversation. Bot collects required information step by step and hands off to the web application for document upload.
Document upload	Document upload is redirected to the web application. Teams bot cannot receive file attachments for AI processing—this is a platform constraint, not a solution limitation.
IT Security requirement	Teams bot integration must pass Montefiore IT Security review. Bot is registered in Azure Active Directory and operates within Montefiore's Microsoft 365 tenant boundary. No data transits external services without encryption and explicit authorization.

Recommended for	Notification delivery, quick-action approvals for managers, status inquiry, and HR alert routing. Strongest as a complement to the web application rather than a standalone channel.

5.4 Integration Capability Comparison

Capability	Web Application	Workday Integration	MS Teams Bot
Full application intake	Full feature set	Full within Workday UX	Guided conversational only
Document upload & AI validation	Full drag-and-drop + AI panel	Within Workday UX	Redirects to web app
Real-time eligibility check	Inline on form	Inline in Workday	Via bot query
AI Agent Output Panel (HR)	Rich panel with all agent data	Summary view	Alert + link to web app
Manager approval	Full approval UI	Workday approval task	Adaptive Card in DM
Payroll posting	Via Payroll Agent API	Direct Workday API write	Via Payroll Agent API
Real-time status tracking	Full status timeline	Workday status widget	Bot query response
NYSNA compliance dashboard	Full dashboard	Summary in Workday	Alert notifications only
Scholarship committee scoring	Full scoring portal	Not supported in Workday	Not supported in Teams
Audit log & reporting	Full interactive log + export	Summary reports	Not available in Teams
Mobile-friendly	Yes (responsive)	Via Workday mobile app	Yes (Teams mobile app)
Recommended primary use	All users; all programs	Employee intake + payroll	Approvals + notifications

7. Deployment Options

Deployment Options

The MTRA can be deployed in three configurations, each with distinct implications for infrastructure ownership, data residency, operational control, and IT governance. All three options meet the compliance requirements specified in Section 7. The appropriate option for Montefiore will be determined through IT Security engagement as confirmed in the discovery session.

6.1 Option 1: Software as a Service (SaaS)

ThinkMindLabs operates the full MTRA platform on a managed cloud infrastructure (AWS GovCloud or Azure Government—both FedRAMP Authorized). Montefiore connects via secured API and SSO federation. No infrastructure management required from Montefiore IT.

SaaS Key Characteristics

- Hosted on HIPAA-eligible cloud infrastructure (AWS GovCloud / Azure Government) with BAA in place
- ThinkMindLabs responsible for all infrastructure operations, patching, scaling, and uptime (SLA: 99.9%)
- Montefiore data remains encrypted at rest (AES-256) and in transit (TLS 1.3); encryption keys held by Montefiore
- All AI model inference performed within the HIPAA-eligible boundary; no PHI or PII sent to general-purpose LLM APIs
- SOC 2 Type II audit reports available to Montefiore on request; penetration test results shared annually
- Fastest time-to-value; no Montefiore infrastructure provisioning required
- Tenant isolation: Montefiore's data is logically isolated in a dedicated tenant; never commingled with other clients

6.2 Option 2: Platform as a Service (PaaS)

The MTRA agent framework is deployed onto cloud infrastructure provisioned and owned by Montefiore within its own Azure or AWS environment. ThinkMindLabs provides the agent software as licensed containers (Kubernetes/Docker); Montefiore IT operates the underlying platform.

PaaS Key Characteristics

- All application data remains within Montefiore's own cloud subscription and data perimeter
- Montefiore IT retains full control of infrastructure configuration, network policies, and cloud security posture
- ThinkMindLabs responsible for agent software updates, AI model updates, and application-layer support
- Montefiore responsible for platform operations: Kubernetes cluster management, cloud networking, certificate management
- Suitable for organizations with existing cloud engineering capability in Azure or AWS
- Enables use of Azure OpenAI Service or AWS Bedrock for LLM inference—both HIPAA-eligible with BAA
- Higher infrastructure ownership for Montefiore IT; greater control over the full technology stack

6.3 Option 3: On-Premise Deployment

The MTRA is deployed entirely within Montefiore's own data centers, with no cloud dependency. All compute, storage, networking, and AI inference runs on Montefiore-owned and operated infrastructure. This option is suitable if Montefiore's IT Security policy prohibits cloud-hosted AI processing.

On-Premise Key Characteristics

- All data, AI inference, and processing remains entirely within Montefiore's physical and network perimeter
- LLM inference runs on on-premise GPU infrastructure or private model deployment (e.g., Llama, Mistral, or enterprise-licensed GPT-4 equivalent via on-premise deployment)
- Maximum data sovereignty and control; no external network dependency for core agent functions
- Montefiore IT fully responsible for infrastructure provisioning, scaling, patching, and availability
- ThinkMindLabs provides software, AI model packages, and ongoing application support via remote access
- Highest operational complexity and infrastructure investment; longest initial setup
- Appropriate for organizations with mature on-premise infrastructure and strict data residency requirements

6.4 Deployment Option Comparison

Dimension	SaaS	PaaS	On-Premise
Data residency	HIPAA-eligible cloud (BAA in place)	Montefiore's own cloud subscription	Montefiore's own data centers
Infrastructure ownership	ThinkMindLabs-managed	Montefiore IT (platform) + ThinkMindLabs (app)	Montefiore IT (full stack)

LLM inference boundary	HIPAA-eligible cloud endpoint	Azure OpenAI / AWS Bedrock (BAA)	On-premise GPU / private model
Montefiore IT effort	Minimal (SSO + API firewall)	Moderate (K8s operations)	High (full stack operations)
Security audit scope	ThinkMindLabs SOC 2 + BAA	Shared: cloud posture + app layer	Full internal security review
Scalability	Elastic (auto-scales)	Elastic (cloud auto-scales)	Constrained by on-prem capacity
AI model updates	Managed by ThinkMindLabs	Managed by ThinkMindLabs	Coordinated deployment required
Disaster recovery	Multi-AZ; RTO < 1 hour	Per Montefiore cloud DR policy	Per Montefiore DR policy
Recommended for	Fastest value; minimal IT burden	Cloud-forward; IT control preference	Strict data sovereignty requirement

6.5 ThinkMindLabs Recommendation

Based on Montefiore's confirmed use of Workday and ServiceNow (both cloud-based), and the stated openness to cloud-based tooling, ThinkMindLabs recommends the SaaS deployment model as the primary option, with PaaS as an alternative if Montefiore IT requires direct infrastructure control. The SaaS model delivers the fastest path to automation while meeting all applicable healthcare compliance requirements through the execution of a Business Associate Agreement and deployment on HIPAA-eligible cloud infrastructure.

Montefiore's IT Security team engagement—confirmed as a requirement in the discovery session—will be initiated in Phase 1 and will inform the final deployment decision. ThinkMindLabs will provide all required security documentation (SOC 2 Type II report, penetration test summary, BAA draft, data flow diagrams, and encryption specification) to support Montefiore IT Security review.

8. Software Compliance — U.S. Healthcare

Software Compliance — U.S. Healthcare

Building software for a U.S. health system carries obligations that extend well beyond standard enterprise software. The MTRA architecture has been designed from the ground up to meet the compliance requirements applicable to Montefiore as a HIPAA-covered entity and New York-regulated healthcare employer. The following section details each applicable standard and how the MTRA satisfies it.

7.1 Compliance Standards Matrix

Regulation / Standard	Applicability	How the Solution Complies
HIPAA Privacy Rule(45 CFR §164.502–164.530)	Applies: employee health benefit data, HR records containing PHI adjacent data	Minimum necessary access enforced. Role-based access control limits each user to only the data required for their function. HR staff cannot access employee health data through the MTRA. PHI is not stored in the tuition system—employee records are referenced by employee ID only.
HIPAA Security Rule(45 CFR §164.304–164.318)	Applies: all electronic Protected Health Information (ePHI) and sensitive employee data in the system	Technical safeguards: AES-256 encryption at rest; TLS 1.3 in transit; MFA enforced for all user access; audit logs for all data access events. Administrative safeguards: access provisioning and de-provisioning automated via SSO; workforce training required. Physical safeguards: cloud infrastructure with physical security controls (AWS GovCloud / Azure Government).
HIPAA Breach Notification Rule(45 CFR §164.400)	Applies: any unauthorized access to or disclosure of PHI	Automated anomaly detection in Audit Agent. Security incident response plan maintained. Breach notification procedure documented and tested. Business Associate Agreement (BAA) executed between Montefiore and ThinkMindLabs before go-live.
SOC 2 Type II(AICPA Trust Services Criteria)	Industry best practice for SaaS and PaaS vendors serving healthcare	ThinkMindLabs MTRA platform audited annually by independent CPA firm against Security, Availability, Confidentiality, and Processing Integrity trust service

		criteria. Audit report available to Montefiore under NDA.
NIST SP 800-53 Rev. 5	Federal security control framework; adopted by many healthcare systems as baseline	MTRA controls mapped to NIST 800-53 control families: AC (Access Control), AU (Audit and Accountability), IA (Identification and Authentication), SC (System and Communications Protection), SI (System and Information Integrity). Control mapping documentation provided for Montefiore IT Security review.
IRS §127 Educational Assistance	Applies: tax treatment of tuition reimbursements above annual exclusion limit ($5,250)	Tax Compliance function in Eligibility Agent automatically calculates taxable vs. non-taxable portion for all reimbursements. W-2 impact recorded. Annual reconciliation report generated. Program 4 (dependent tuition for executives) specifically designed for correct gross-up and withholding treatment.
NYSNA Collective Bargaining Agreement	Applies: all NYSNA-covered nurse applicants under Program 1	NYSNA policy rules hard-coded as non-negotiable constraints: 45-day reimbursement window, service agreement for amounts > $3,000, nursing-related program requirement. Any changes to NYSNA rules require Labor Relations review and configuration update. NYSNA-specific compliance report generated automatically.
New York SHIELD Act(NY General Business Law §899-aa)	Applies: protection of private information of New York residents	Reasonable security program implemented covering administrative, technical, and physical safeguards. Data inventory maintained. Employee PII handled per SHIELD Act requirements. Breach notification procedures compliant with NY SHIELD 72-hour notification requirement.
ADA / Section 504 Accessibility	Applies: web application and digital tools used by employees	Web application built to WCAG 2.1 Level AA. Screen reader compatibility (ARIA labels). Keyboard navigation for all functions. Color contrast meeting minimum 4.5:1 ratio. Accessibility audit performed before go-live.
Equal Employment Opportunity(Title VII, ADA, ADEA)	Applies: scholarship and tuition benefit administration must be non-discriminatory	AI scoring for scholarship program audited for disparate impact across protected classes before deployment. All AI-assisted decisions have human override capability. Audit Agent monitors exception

		patterns for potential bias indicators. Annual bias audit conducted by independent reviewer.
21st Century Cures Act / Interoperability	Applicable for health systems using EHR data—limited applicability here; included for completeness	MTRA does not access or process EHR clinical data. Employee benefit data is distinct from clinical records. FHIR-compatible API design ensures interoperability if future integration with clinical systems is desired.
FedRAMP (where applicable)	Applies to cloud infrastructure hosting; relevant for SaaS and PaaS deployment	SaaS deployment on AWS GovCloud (FedRAMP High Authorized) or Azure Government (FedRAMP High Authorized). PaaS deployment on Montefiore's own FedRAMP-authorized cloud subscription. FedRAMP authorization certificates available on request.

7.2 Business Associate Agreement (BAA)

As a condition of deployment and data processing on behalf of Montefiore Health System—a HIPAA Covered Entity—ThinkMindLabs will execute a Business Associate Agreement (BAA) with Montefiore prior to any system go-live. The BAA governs the use, disclosure, and safeguarding of Protected Health Information and sensitive employee benefit data. ThinkMindLabs' standard BAA template is available for Montefiore Legal review. Additionally, BAAs are in place with all sub-processors, including cloud infrastructure providers (AWS, Microsoft Azure) and any third-party AI inference providers used within the solution boundary.

7.3 AI Governance Framework

As an organization that confirmed the need to engage IT Security and align to department-wide AI governance rules, Montefiore will require a structured AI governance framework for the MTRA. ThinkMindLabs proposes the following governance structure:

AI Review Board	Cross-functional committee (HR, Legal, Compliance, IT Security, DEI) meeting quarterly to review AI agent decision patterns, exception logs, and bias audit results. ThinkMindLabs provides quarterly AI performance report as input.
Explainability standard	Every AI agent decision generates a human-readable explanation stored in the case record. No AI decision is presented as a black box. HR staff can access the full reasoning chain for any case at any time.

Human override requirement	All AI auto-approvals and auto-denials can be overridden by authorized HR staff. Override reason is logged. Override patterns are reviewed quarterly by the AI Review Board.
Bias monitoring	Annual independent bias audit of AI eligibility and scoring decisions across protected classes (race, gender, ethnicity, age, disability status, union membership). Audit results reported to AI Review Board and HR Director.
Model update governance	All AI model updates (LLM version changes, ruleset updates, scoring model retraining) subject to a change management process including UAT and HR sign-off before production deployment.
Employee transparency	Employees are notified that AI agents assist in processing their applications. Plain-language explanation of AI's role available in the portal. Employees have the right to request human review of any AI-assisted decision within 5 business days.
IT Security alignment	MTRA architecture documentation, data flow diagrams, AI model specifications, and third-party processor list provided to Montefiore IT Security for review prior to deployment. Ongoing annual security review included in support agreement.

9. Data Architecture & Integration

Data Architecture & Integration

8.1 Data Flow Overview

All data flows within the MTRA follow a zero-trust architecture. No data element moves between systems without explicit authentication, authorization, and encryption. The following describes the primary data flows that power the agent ecosystem.

Workday HRIS → MTRA	REST API read integration. Pulls: employee tenure, FTE status, job classification, union code, dependent records, active employment status. Called at intake (eligibility check) and on departure events (SA compliance). Read-only access; MTRA cannot modify Workday HRIS records.
MTRA → Workday Payroll	REST API write integration. Pushes: approved reimbursement amount, employee ID, GL cost center code, tax classification, payment reference. Write access limited to approved payroll posting endpoint only. All writes logged.
MTRA — ServiceNow	Bidirectional REST API. Reads and writes case records, attachments, notifications, and approval tasks. ServiceNow remains the system of record for case management. MTRA agents update case status, attach documents, and log actions in ServiceNow.
MTRA — DocuSign / Adobe Sign	REST API for signature request creation, status polling, and signed document retrieval. Signed documents returned to MTRA and stored in employee record in ServiceNow / SharePoint. Signature events logged with timestamp and signatory identity.
MTRA — SharePoint / S3	Document storage integration. All uploaded documents stored in Montefiore-controlled document store with metadata tagging (case ID, program type, document type, upload date, extraction status). MTRA agents reference documents by secure URL; document content processed in-memory and not persisted in MTRA database.
MTRA — IPEDS / DOE	Read-only API query for institutional accreditation verification. Called on submission of Program 4 (dependent tuition) cases. Query uses institution name; no employee PII is transmitted to the DOE database.

AI Inference Boundary	LLM inference calls (document extraction, compliance reasoning, notification drafting) processed within HIPAA-eligible boundary. Employee identifiers replaced with tokenized case IDs in all LLM prompts. PII is not included in model inputs; extracted from outputs post-inference.

8.2 Data Retention

In alignment with the client's confirmed requirement of 10-year data retention, the MTRA implements the following retention schedule:

Case records (all programs)	10 years from case closure. Includes application data, agent decision logs, approval records, payment confirmation.
Supporting documents	10 years from upload. Transcripts, receipts, acceptance letters, CME certificates, service agreements, dependent verification documents.
Audit log (agent + human actions)	10 years. Immutable; cannot be modified or deleted by any user. Accessible to HR Director, Internal Audit, and Legal on request.
Payroll posting records	7 years minimum (IRS requirement); 10 years per Montefiore policy. Includes amount, tax treatment, and GL entry.
Service agreements	10 years from expiration of commitment period. Required for potential NYSNA grievance or repayment dispute resolution.
Notification logs	5 years. Record of all communications sent to employees, managers, and HR including timestamp, channel, and content.
AI decision explanations	10 years. Co-located with case record. Required for compliance, bias audit, and employee rights to review.

10. Expected Outcomes & Success Metrics

Expected Outcomes & Success Metrics

The following outcomes are projected based on benchmarks from comparable U.S. health system AI workflow implementations and the specific process data confirmed in the Montefiore discovery session (approximately 1,000 cases per year; >50% requiring multiple follow-ups; current processing times extending well beyond the next-paycheck goal).

Metric	Current State	Target State (MTRA)	Improvement
Applications requiring multiple follow-ups	>50% of 1,000 cases	<5% of cases	**~90% reduction**
HR processing time per case (clean submission)	4–6 hours	< 1 hour (exception review only)	**~85% reduction**
End-to-end processing time	15–25 business days	3–5 business days	**~80% reduction**
Payment timeline after docs received	Inconsistent; multi-cycle	Next paycheck cycle	**Target met**
NYSNA 45-day compliance rate	Manual tracking; inconsistent	>99% (automated enforcement)	**Near-perfect compliance**
Service agreement on-file rate	Manual; gaps estimated 15–25%	>99% (automated generation & routing)	**~90% improvement**
18-credit cap exception rate	High (top confusion driver)	Near-zero (proactive balance alerts)	**~95% reduction**
Duplicate or over-limit submissions	Undetected; estimated 3–5%	<0.5% (AI detection at intake)	**~90% reduction**
Audit preparation time	40–60 hrs per audit	<5 hours (dashboard-driven)	**~90% reduction**
Employee satisfaction with benefit process	Low (all aspects cited as difficult)	Target: 80%+ satisfaction score	**Significant improvement**

HR FTE hours on tuition admin (annual)	~10,000–15,000 hrs/year	~2,000–3,000 hrs/year	**~80% reduction**

9.1 Success Criteria Aligned to Client Definition

The client defined successful modernization as: "Remove manual and automate it all — remaining consistent, building strong internal cadence and control." The MTRA addresses this definition across three dimensions:

Remove Manual — What Gets Automated

- Document reading, field extraction, and validation: 100% automated for standard submissions
- Eligibility checks and credit/dollar balance calculations: 100% automated at point of intake
- Missing document detection and employee follow-up: 100% automated with personalized outreach
- Approval routing, reminder dispatch, and escalation: 100% automated
- Service agreement generation, e-signature routing, and lifecycle monitoring: 100% automated
- Payroll posting with correct tax treatment: 100% automated on final approval
- Audit logging and compliance reporting: 100% automated and continuously maintained

Strong Internal Cadence — What Gets Enforced Consistently

- NYSNA 45-day window: monitored continuously; HR and NYSNA Compliance Officer alerted automatically
- Service agreement threshold ($3,000): enforced at intake; no case progresses without SA on file
- 18 credits per year cap: checked at intake; employee shown current balance; over-limit submissions blocked
- Approval chain completeness: all required signatures tracked; no case marked approved without full chain
- Communication standards: all employee-facing communications rewritten and consistent; no ad-hoc HR emails

Long-Term Scalability — Agent Framework Is Extensible

- New tuition programs can be added by configuring a new program definition and associated policy rules—no code changes required
- Policy rule updates (e.g., NYSNA CBA renegotiation, IRS exclusion limit changes) managed by HR Administrators in a policy configuration UI
- Additional AI agent capabilities (e.g., predictive submission reminders, fraud detection, integration with credentialing systems for CME) can be added to the agent registry incrementally
- Agent performance and utilization metrics reviewed with Montefiore quarterly; roadmap updated based on operational learnings

11. Why This Approach for Montefiore

Why This Approach for Montefiore

The MTRA is not a generic automation platform adapted for healthcare. It is a purpose-designed agentic AI system built around the specific programs, policy constraints, union obligations, and workforce characteristics that define Montefiore's tuition benefit ecosystem.

The discovery session confirmed that Montefiore's primary challenge is not a technology gap—ServiceNow and Workday are already in place. The challenge is that the current tools require humans to perform work that AI agents can do faster, more consistently, and with a complete audit trail: reading documents, checking rules, chasing signatures, and re-keying data between systems. The MTRA inserts AI agents precisely into those gaps, leaving human judgment for the decisions that genuinely require it.

10.1 Healthcare-Specific Considerations

- The MTRA treats compliance as a first-class design requirement—not an afterthought. HIPAA, SOC 2, NIST, NYSNA CBA, IRS §127, and New York SHIELD Act requirements are embedded in the architecture.
- The AI governance framework respects that Montefiore is in a learning phase with AI—the system is designed to build confidence through explainability, human override, and transparent audit trails before expanding automation boundaries.
- The agent framework is union-aware. NYSNA rules are encoded as non-negotiable constraints. The system is designed so that CBA renegotiation results in a configuration update, not a development project.
- Physical Therapist CME—identified in discovery as requiring union rule review—is handled with an explicit compliance flag and Labor Relations routing step. No union obligation is automated away without deliberate authorization.

10.2 Alignment with Montefiore’s Strategic Mission

Montefiore’s mission—to heal, teach, discover, and advance health—depends on a workforce that feels valued and supported. Tuition reimbursement is a tangible expression of that commitment. When the benefit experience is frustrating, slow, and opaque, the benefit’s retention value is undermined. When it is fast, transparent, and frictionless, it reinforces Montefiore’s reputation as an employer of choice in one of the most competitive healthcare talent markets in the United States.

Every hour of HR staff time recovered from manual document processing is an hour available for the higher-value work that defines Montefiore's people operations: career development conversations, workforce equity initiatives, and strategic HR partnership with clinical leadership. The MTRA does not replace HR—it gives HR the capacity to do more of what matters.

ThinkMindLabs Commitment to Montefiore

- Deep healthcare compliance expertise: every design decision evaluated against HIPAA, SOC 2, NIST, and New York State requirements
- Agentic AI experience: not a generic workflow tool—a purpose-built multi-agent system with healthcare-specific domain knowledge
- ServiceNow + Workday native integration: no disruption to existing systems; agents augment the platforms already in use
- Union-aware design: NYSNA and applicable PT union rules treated as architectural constraints, not afterthoughts
- Transparent AI governance: explainability, human override, and bias monitoring built in from day one
- Long-term partnership model: quarterly business reviews, roadmap collaboration, and continuous improvement as Montefiore's AI maturity grows

Montefiore Health System | Intelligent Tuition Reimbursement Agent

Prepared by ThinkMindLabs | April 2026 | Version 1.0

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